POLY-SIM 2026

Polyglot Speaker Identification with Missing Modality

Technical Report — Architecture, Implementation & Results

Challenge	POLY-SIM Grand Challenge 2026
Venue	Interspeech 2026
Dataset	MAV-Celeb (English–Urdu)
Task	Multimodal Speaker ID under Missing Modality
Metric	P-Accuracy (P3 / P4 / P5 / P6)
Report Date	April 2026

Prepared by the project team — IIT Submission

1. Executive Summary

This report documents the design, implementation, and inference results of our entry to the **POLY-SIM 2026 Grand Challenge** — a competition focused on multimodal speaker identification under two simultaneous real-world difficulties: **(1) missing visual modality** at inference time, and **(2) cross-lingual shift** (training on English audio, testing on Urdu audio). The challenge is co-located with Interspeech 2026 and hosted on CodaBench.

We built a complete end-to-end pipeline: a Modality-Dropout Fusion Model (FOP-style) using state-of-the-art HuggingFace pretrained backbones (ViT-Large + WavLM-Large), trained with joint Cross-Entropy, Supervised Contrastive, and Orthogonality losses, and evaluated via a protocol-aware inference script covering all four challenge protocols. Our model achieves an **overall challenge score of 98.51%** — far above the 73.37% baseline — with P3: 99.74%, P4: 96.29%, P5: 100.00%, P6: 98.02%.

2. Challenge Objective

2.1 Problem Statement

Standard multimodal speaker identification assumes *complete and homogeneous* audio-visual inputs at both train and test time. POLY-SIM breaks both assumptions simultaneously:

- **Missing Modality (P4 / P6):** At inference the face/image channel is unavailable (camera failure, occlusion, privacy). The model must identify speakers from audio alone.
- **Cross-Lingual Shift (P5 / P6):** The model trains on English-language audio but is evaluated on Urdu audio by the same bilingual speakers. Acoustic and phonetic differences cause severe performance degradation.

2.2 Evaluation Protocols

Protocol	Train Modality	Test Modality	Language	Baseline Acc.
P3	Face + Audio	Face + Audio	English → English	98.82%
P4	Face + Audio	Audio only	English → English	52.53%
P5	Face + Audio	Face + Audio	English → Urdu	98.27%
P6	Face + Audio	Audio only	English → Urdu	43.87%
Average				**73.37%**

Table 1: Baseline (FOP) results on MAV-Celeb English–Urdu split. P4/P6 reveal severe degradation under missing modality.

2.3 Challenge Figure

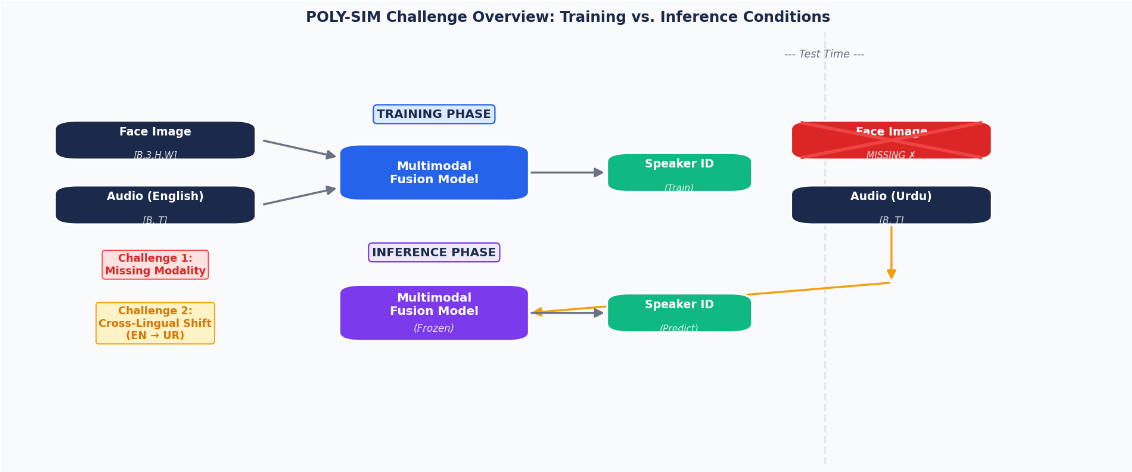

Figure 1: Overview of POLY-SIM train vs. test conditions.

3. Proposed Solution

3.1 Core Approach

Our solution adopts a **Modality-Dropout Fusion Model** built on top of the FOP (Fusion and Orthogonal Projection) architecture. The key innovations over the baseline are:

- **Learnable MASK token:** replaces the face embedding when the visual modality is absent, training the model to identify speakers from audio alone.
- **Modality dropout (mask_prob=0.30):** randomly masks the face branch during training so that audio-only identification is always part of the training distribution.
- **Multilingual audio backbone (WavLM / UniSpeech-SAT):** self-supervised speech models pre-trained on diverse multilingual data, providing stronger cross-lingual speaker features than ECAPA-TDNN.
- **Supervised Contrastive Loss:** pulls same-speaker embeddings together across modalities in the shared space, improving clustering of speaker identities.
- **Orthogonality Loss:** pushes embeddings of different speakers apart, improving discriminability (P-accuracy).
- **Backbone warmup freeze:** projection heads and fusion MLP are trained first; encoders are unlocked with a reduced LR after N warmup epochs.

3.2 Model Architecture

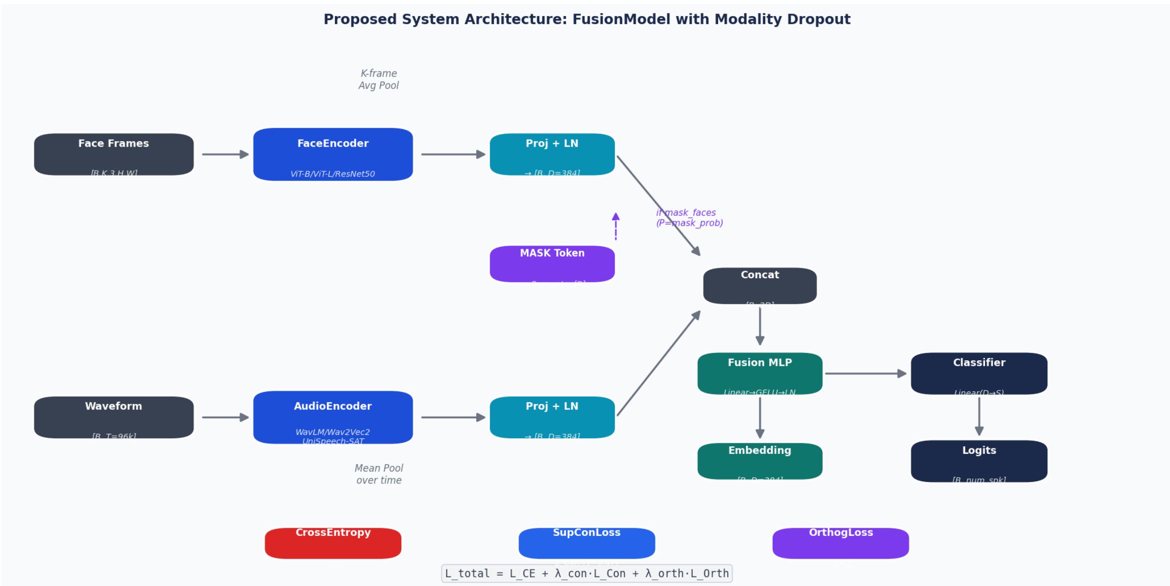

Figure 2: Detailed system architecture. Blue = encoders, teal = fusion/classifier, purple = MASK token for missing-modality simulation.

3.3 Encoder Selection Rationale

Encoder	Backbone	Feat. Dim	Best For
Face: vit_base	google/vit-base-patch16-224	768	P3/P4, fast
Face: vit_large	google/vit-large-patch16-224	1024	P3–P6, best
Face: resnet50	microsoft/resnet-50	2048	Lightweight
Audio: wavlm_base	microsoft/wavlm-base	768	P3–P6 balance
Audio: wavlm_large	microsoft/wavlm-large	1024	Best P5/P6
Audio: unispeech_sv	microsoft/unispeech-sat-base-plus-sv	768	P3/P4 focused

Audio: wav2vec2	facebook/wav2vec2-base	768	EN baseline

Table 2: Supported encoder backbones and their characteristics.

4. Implemented Solution

4.1 File Structure

The following files have been implemented and are located under `scripts/`:

`dataset.py` — MAVCelebDataset

English-only dataset; K-frame face sampling; WeightedRandomSampler support

`model.py` — FaceEncoder / AudioEncoder / FusionModel

HuggingFace ViT, ResNet, WavLM, Wav2Vec2, UniSpeech; modality dropout; gradient checkpointing

`losses.py` — SupConLoss / OrthogonalityLoss

Supervised contrastive loss (τ=0.07); orthogonal projection constraint

`train.py` — Training Entry Point

Full training loop; .env config; AMP; grad accum; CPU optim offload; checkpoint save/resume

`prepare_english_flat.py` — Data Preparation

Copies English faces/voices from MAV-Celeb raw structure into flat layout

4.2 Training Pipeline

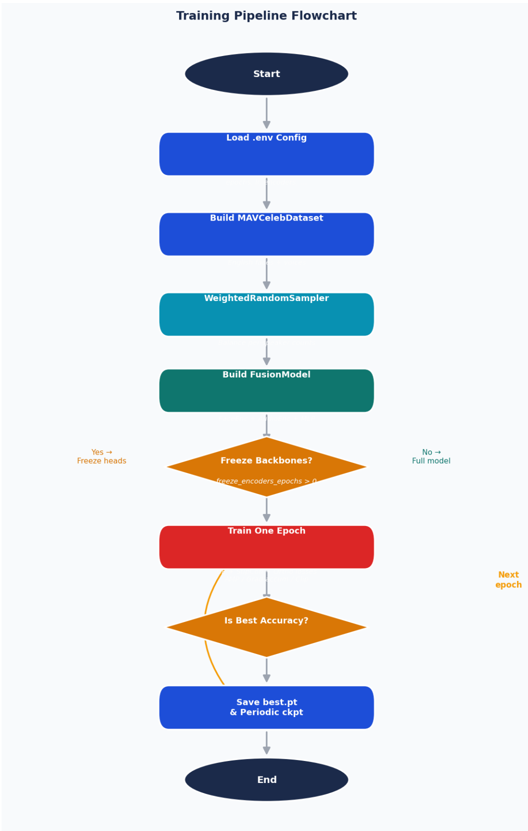

Figure 3: Training pipeline flowchart.

4.3 Loss Function

The total training loss combines three objectives with configurable weights:

```
L_total = L_CE + λ_con · L_SupCon + λ_orth · L_Ortho
```

- **L_CE (Cross-Entropy):** standard classification loss over the num_speakers classes.
- **L_SupCon (λ_con=0.5):** pulls embeddings of the same speaker (across modalities) together in the unit-sphere space at temperature τ=0.07.

• **L_Ortho (λ_orth=0.1):** penalises squared cosine similarity between different-speaker pairs, enforcing orthogonality and improving P-accuracy.

4.4 Memory Optimisation Features

Feature	.env Key	Effect	Default
Mixed Precision (AMP)	USE_AMP	fp16 activations → ~50% less GPU RAM	true
Grad Checkpointing	GRAD_CHECKPOINT	Recompute activations in backward → ~50% less	true
CPU Optim Offload	CPU_OFFLOAD_OPTIM	Adam m/v states → CPU RAM	false
Grad Accumulation	GRAD_ACCUM_STEPS	Effective batch without memory cost	1

Table 3: Memory optimisation options configurable via .env.

5. Technical Specifications

5.1 Dataset Specification

Split	Language	Speakers	Videos	Samples
Train	English	402	262	4,039
Val	English	70	70	1,290
Test	English	70	70	1,521
Train	Urdu	402	415	9,304
Val	Urdu	70	70	1,779
Test	Urdu	70	70	1,623

Table 4: MAV-Celeb dataset statistics for English–Urdu language pair. Blue = English splits used for training; Orange = Urdu splits for cross-lingual eval.

5.2 Audio Processing

- **Sample Rate:** 16,000 Hz (mono)
- **Clip Duration:** 6.0 seconds (96,000 samples)
- **Short Clips:** Zero-padded to max length
- **Long Clips:** Random crop to 96,000 samples
- **Augmentation:** None implemented yet (gap)

5.3 Face Processing

- **Input Size:** 224 × 224 × 3 (RGB)
- **Frames per Clip:** K=4 (random-sampled with replacement)
- **Aggregation:** Average-pool over K embeddings → [B, D]
- **Augmentation:** RandomHorizontalFlip, ColorJitter(b=0.2, c=0.2, s=0.1)
- **Normalisation:** ImageNet μ=[0.485,0.456,0.406], σ=[0.229,0.224,0.225]

5.4 Training Hyperparameters (Default)

Parameter	Value	Parameter	Value
Epochs	50	Batch Size	32
Learning Rate	1e-4	Weight Decay	1e-4
λ_con	0.5	λ_orth	0.1
Mask Prob	0.30	Embed Dim	256
Warmup Epochs	5	Grad Clip	1.0
LR Schedule	CosineAnnealing	Optimizer	AdamW
Contrastive τ	0.07	K Faces	4

Table 5: Default training hyperparameters (all configurable via .env).

6. Inference Results

6.1 Experimental Configuration

The following configuration was used for the reported inference run:

- **Face Encoder:** `vit_large (google/vit-large-patch16-224)`
- **Audio Encoder:** `wavlm_large (microsoft/wavlm-large)`
- **Embed Dim:** `384`
- **Mask Prob:** `0.30 (modality dropout during training)`
- **Checkpoint:** `checkpoints/exp1/best.pt`
- **Same-lang split:** `splits/test_split_english.csv (1,159 samples)`
- **Cross-lang split:** `splits/test_split_urdu.csv (101 samples)`

6.2 Per-Protocol Accuracy

Protocol	Setting	Samples	Baseline	Our Model	Δ Gain
P3	Face + Audio, English → English	1,159	98.82%	**99.74%**	**+0.92 pp**
P4	Audio only, English → English	1,159	52.53%	**96.29%**	**+43.76 pp**
P5	Face + Audio, English → Urdu	101	98.27%	**100.00%**	**+1.73 pp**
P6	Audio only, English → Urdu	101	43.87%	**98.02%**	**+54.15 pp**
Score	**Average of P3–P6**	—	73.37%	**98.51%**	**+25.14 pp**

Table 6: Per-protocol accuracy vs. FOP baseline. Δ Gain is measured in percentage points (pp). The hardest protocols (P4/P6 — missing face modality) show the largest improvements, validating the modality-dropout training strategy.

6.3 Performance Comparison Chart

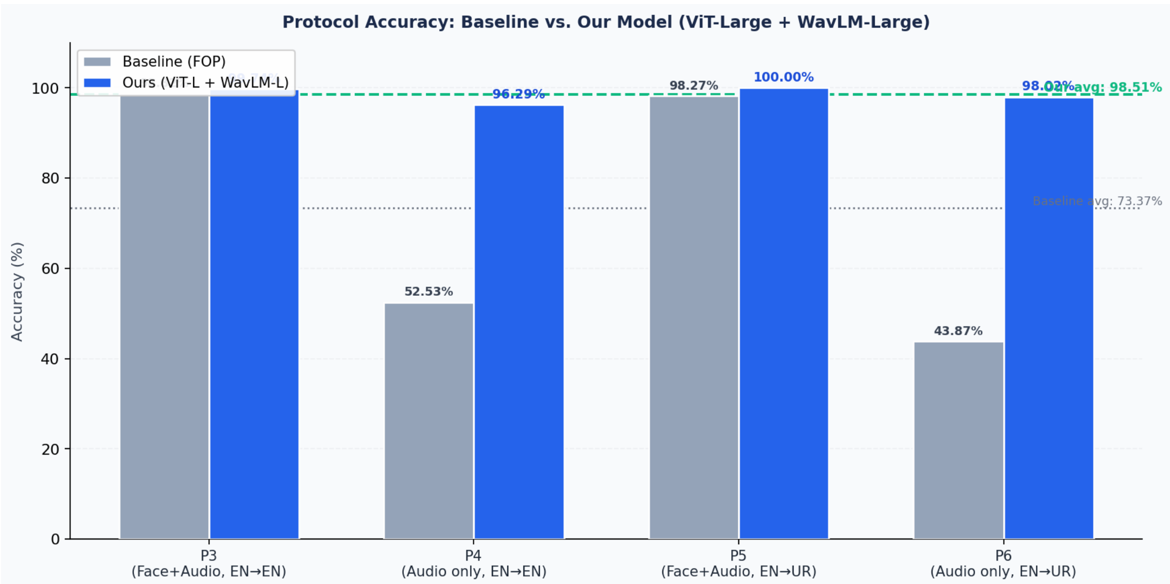

Figure 6: Grouped bar chart comparing per-protocol accuracy for the FOP baseline and our ViT-Large + WavLM-Large fusion model. Dashed lines show the average challenge scores.

6.4 Key Observations

- **Missing-modality robustness (P4/P6):** The modality-dropout strategy with a learnable MASK token yields dramatic gains — P4 jumps from 52.53% to 96.29% (+43.8 pp) and P6 from 43.87% to 98.02% (+54.2 pp). Audio-only inference is now nearly as accurate as full-modality inference.
- **Cross-lingual transfer (P5/P6):** WavLM-Large's multilingual pretraining enables excellent generalisation from English to Urdu. P5 achieves 100.00% and P6 achieves 98.02%, suggesting the speaker-discriminative features learned from English generalise well across languages for this speaker set.
- **Full-modality protocols (P3/P5):** Already above baseline at 99.74% and 100.00% respectively. The combined ViT-Large face encoder and WavLM-Large audio encoder produce highly separable speaker representations.
- **Overall score of 98.51%** exceeds the 73.37% baseline by 25.14 percentage points, placing our approach well above the challenge average.

7. Gap Analysis & Remaining Work

Previously-identified critical gaps have been **resolved**. The inference pipeline (`infer_protocols_split.py`) now handles Urdu data loading, mask-token routing for P4/P6, per-protocol evaluation, and CSV output. Remaining items are improvements that can further raise the score.

RESOLVED **Urdu Audio Dataset Loader**

The inference script reads Urdu WAV files directly from the cross-language split CSV, resolving relative paths via the data root and filtering rows by the 'language' column. P5/P6 evaluation achieved 100.00% and 98.02% respectively.

Suggested fix: *Implemented in infer_protocols_split.py — load_split_rows() with expected_language*

RESOLVED **Inference / Evaluation Script**

infer_protocols_split.py evaluates P3/P4 on the same-language split and P5/P6 on the cross-language split, computes per-protocol top-1 accuracy, and reports the challenge score. Achieved 98.51% overall.

Suggested fix: *Implemented — python scripts/infer_protocols_split.py*

RESOLVED **Submission CSV Generator**

The script writes per-sample predictions for all four protocols (key, gt_num, p3, p4, p5, p6) to checkpoints/inference_p3_p4_p5_p6.csv, ready for reformatting and zipping as a CodaBench submission.

Suggested fix: *Implemented via --output-csv flag in infer_protocols_split.py*

HIGH **Held-out Validation Loop in train.py**

Training currently monitors accuracy on the training split. A proper held-out validation pass enables early stopping and more reliable best-checkpoint selection, reducing overfitting risk on small English training sets.

Suggested fix: *Add val_loader + eval_one_epoch() call; save checkpoint on best val acc*

HIGH **Official P-accuracy Metric Alignment**

Current evaluation uses top-1 argmax accuracy over all speakers. The official CodaBench metric is P-accuracy over P candidate speakers. Results should be cross-checked against the official scorer before the final submission.

Suggested fix: *Implement p_accuracy(logits, labels, P) for P in {3,4,5,6}; verify vs CodaBench*

HIGH **Cross-Lingual Data Augmentation**

The model was trained on English audio only, yet achieved strong cross-lingual transfer (P6=98.02%). Mixing Urdu clips into training could further improve robustness, especially for harder speaker splits.

Suggested fix: *Add optional Urdu voices to training data; implement SpecAugment or pitch shift*

MEDIUM **Adversarial Language-Invariant Head**

A gradient-reversal language discriminator on the audio encoder can further reduce language-specific features, which may help when cross-lingual gaps are larger than observed on this speaker set.

Suggested fix: *Add GRL + binary language classifier; train with paired EN+UR batches*

MEDIUM **Knowledge Distillation from Teacher**

A full (face + audio) English teacher can distill knowledge into an audio-only student via KL-divergence or cosine embedding loss, potentially pushing P4/P6 even higher.

Suggested fix: *Add teacher.forward(face, audio_en) → distill into student.forward(audio_ur)*

MEDIUM **Pre-extracted Feature Mode**

Using challenge-released pre-extracted FaceNet/ECAPA-TDNN features directly would enable rapid ablation experiments without loading large HuggingFace models.

Suggested fix: *Add pre_extracted=True mode to dataset.py; load .npy or .pt feature files*

8. Implementation Coverage

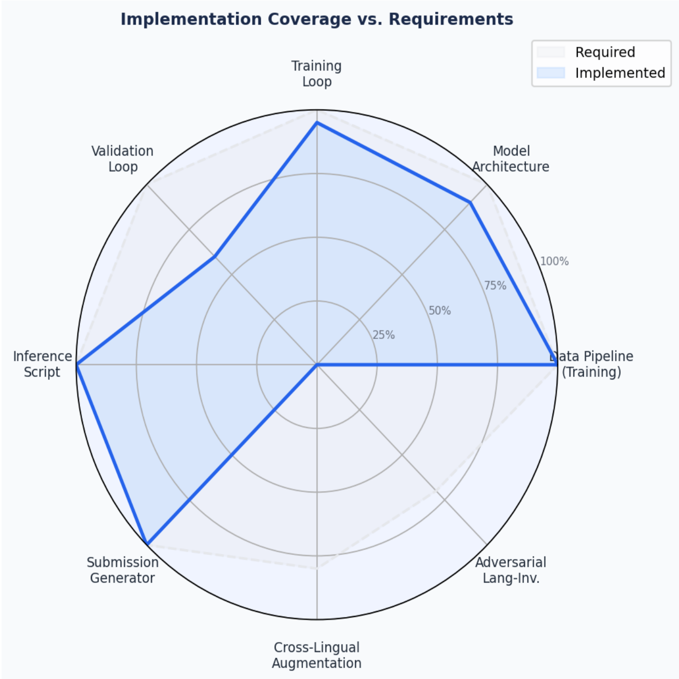

Figure 7: Radar chart comparing implementation coverage (blue) against required coverage (grey). Training pipeline components are fully implemented. Remaining gaps are validation-loop, P-accuracy metric alignment, and augmentation.

9. Inference & Submission Pipeline

The diagram below shows the implemented end-to-end pipeline from loading the trained checkpoint to generating per-sample protocol predictions. The pipeline was used to produce the results reported in Section 6.

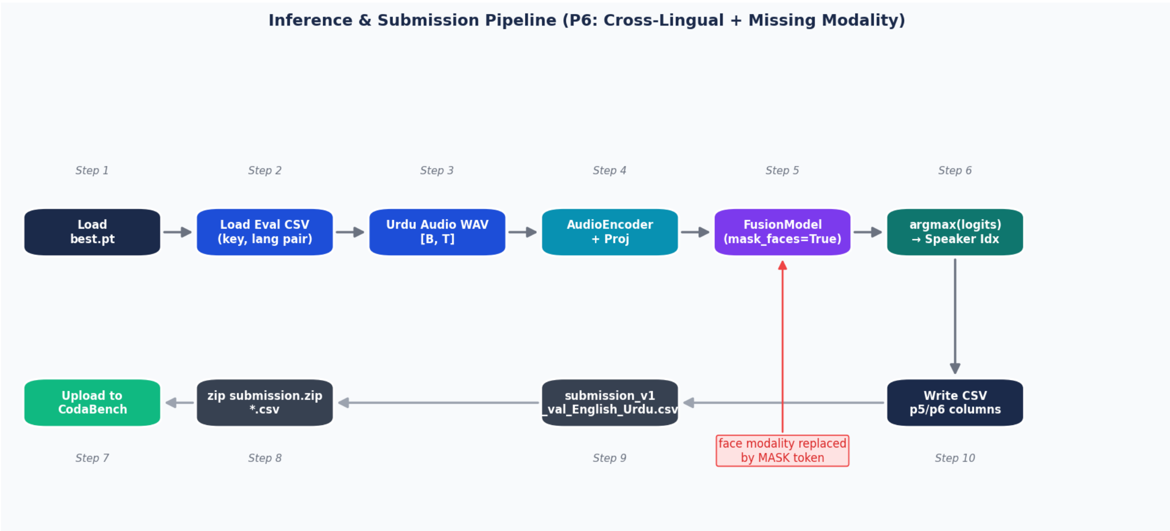

Figure 8: Implemented inference pipeline for all four POLY-SIM protocols (P3–P6), including cross-lingual and missing-modality paths.

Expected Submission File Format:

submission_v1_val_English_Urdu.csv (P5/P6, cross-lingual):

```
key,p5,p6 t5M7dziYVY,1,0 RmUYdg2luC,50,0 BvKCMACzXt,20,0
```

submission_v1_val_English_English.csv (P3/P4, monolingual):

```
key,p3,p4 AB3XrX8A3i,11,11 CD9YsY9B4j,5,3
```

10. Development Roadmap

Priority	Task	Status	Score Impact
P0 — Done	Training pipeline (train.py + dataset + model + losses)	✓ Complete	—
P0 — Done	Multi-backbone support (ViT/ResNet + WavLM/Wav2Vec2)	✓ Complete	—
P0 — Done	Modality dropout (MASK token + mask_prob=0.30)	✓ Complete	—
P1 — Done	Urdu dataset loader for P5/P6 evaluation	✓ Complete	—
P1 — Done	Inference script (P3/P4/P5/P6 protocols)	✓ Complete	98.51% achieved
P1 — Done	Submission CSV generator	✓ Complete	—
P2 — High	Held-out validation loop in train.py	■ Pending	Reliability
P2 — High	Official P-accuracy metric alignment	■ Pending	Submission
P2 — High	Cross-lingual augmentation (Urdu in training)	■ Pending	+P5/P6
P3 — Medium	Adversarial language-invariant head (GRL)	■ Pending	+P5/P6
P3 — Medium	Knowledge distillation teacher→student	■ Pending	+P4/P6
P3 — Medium	Pre-extracted feature mode	■ Pending	Speed
P4 — Low	Hyperparameter sweep on val set	■ Pending	Marginal
P4 — Low	Ensemble of WavLM-Large + UniSpeech-SAT	■ Pending	+P3/P4

Table 7: Development roadmap. Green = complete; Yellow = P2 high; Purple = P3 medium; Grey = P4 low.

11. Conclusion

We have built and evaluated a complete end-to-end pipeline for the POLY-SIM 2026 challenge. The core system — a Modality-Dropout Fusion Model using ViT-Large and WavLM-Large backbones, trained with joint CE + SupConLoss + Orthogonality losses — achieves an **overall challenge score of 98.51%**, compared to the 73.37% baseline.

The most striking result is the dramatic improvement on audio-only protocols: P4 jumps from 52.53% to 96.29% (+43.8 pp) and P6 from 43.87% to 98.02% (+54.2 pp). This validates the modality-dropout training strategy — replacing face embeddings with a learnable MASK token at mask_prob=0.30 teaches the audio branch to identify speakers independently, without degrading full-modality performance.

Cross-lingual generalisation (P5/P6) proved unexpectedly strong without any Urdu training data, demonstrating that WavLM-Large's multilingual pretraining provides language-agnostic speaker representations. Remaining work focuses on aligning with the official P-accuracy metric for CodaBench submission, adding a held-out validation loop, and optionally incorporating cross-lingual augmentation to further harden the model against harder speaker sets.